

PRINCIPLES OF GEOGRAPHIC INFORMATION SYSTEM SEM: VI

SEM VI: UNIT 2



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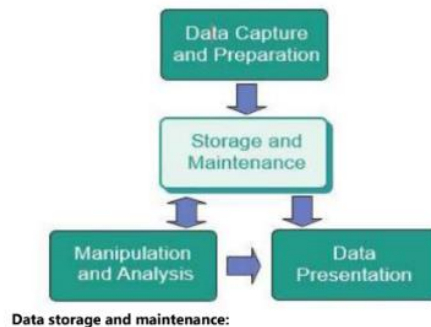
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1. List the functional components of GIS. Explain any two of them in details.
(Apr-19)(Nov-19)

Ans:

A GIS consists of several functional components—components which support key GIS functions. These are data capture and preparation, data storage, data analysis, and presentation of spatial data. Figure shows a diagram of these components, with arrows indicating the data flow in the system. For a particular GIS, each of these components may provide many or only a few functions.



Data management refers to the storage and maintenance of the data transmitted by the buoys via satellite communication. This phase requires a decision to be made on how best to represent our data, both in terms of their spatial properties and the various attribute values which we need to store.

- **The data presentation :**

The data presentation phase deals with putting it all together into a format that communicates the result of data analysis in the best possible way. Many issues arise in this phase. Among other things, we need to consider what the message is that we want to portray, who the audience is, what kind of presentation medium will be used, which rules of aesthetics apply, and what techniques are available for representation.

- **Spatial data capture and preparation:**

The functions for capturing data are closely related to the disciplines of surveying engineering, photogrammetry, remote sensing, and the processes of digitizing, i.e. the conversion of analogue data into digital representations. Remote sensing, in particular, is the field that provides photographs and images as the raw base data from which spatial data sets are derived.

- **Spatial data storage and maintenance**

The way that data is stored plays a central role in the processing and the eventual understanding of that data. In most of the available systems, spatial data is organized in layers by theme and/or scale. For instance, the data may be organized in thematic categories, such as land use, topography and administrative subdivisions, or according to map scale.

- **Spatial query and analysis:**

The most distinguishing parts of a GIS are its functions for spatial analysis, i.e. operators

that use spatial data to derive new geoinformation. Spatial queries and process models play an important role in this functionality. One of the key uses of GISs has been to support spatial decisions. Spatial decision support systems (SDSS) SDSS are a category of information systems composed of a database, GIS software, models, and a so-called knowledge engine which allow users to deal specifically with locational problems.

- **Spatial data presentation:**

The presentation of spatial data, whether in print or onscreen, in maps or in tabular displays, or as 'raw data', is closely related to the disciplines of cartography, printing and publishing. The presentation may either be an end-product, for example as a printed atlas, or an intermediate product, as in spatial data made available through the internet.



2. Explain the various reasons for using DBMS in GIS.

(Apr-19)

OR

What is the reason for using DBMS in GIS?

(Nov-19)

OR

Explain the various reasons for using DBMS in GIS

(Apr-23)

Ans:

-Database management systems:

A database management system (DBMS) is a software package that allows the user to set up, use and maintain a database

-Reasons for using a DBMS:

There are various reasons why one would want to use a DBMS for data storage and processing. A DBMS supports the storage and manipulation of very large data sets. Some data sets are so big that storing them in text files or spreadsheet files becomes too awkward for use in practice. The result may be that finding simple facts takes minutes, and performing simple calculations perhaps even hours. A DBMS is specifically designed for this purpose.

-A DBMS can be instructed to guard over data correctness.

An important aspect of data correctness is data entry checking: ensuring that the data that is entered into the database does not contain obvious errors. For instance, since we know the study area we are working in, we also know the range of possible geographic coordinates, so we can ensure the DBMS checks them.

-A DBMS supports the concurrent use of the same data set by many users

Large datasets are built up overtime, which means that substantial investments are required to create and maintain them, and that probably many people are involved in the data collection, maintenance and processing. These data sets are often considered to be of a high strategic value for the owner(s), which is why many may want to make use of them within an organization.

-A DBMS provides a high-level, declarative query language.

A query is a computer program that extracts data from the database that meet the conditions indicated in the query. A DBMS supports the use of a data model. A data model is a language with which one can define a database structure and manipulate the data stored in it.

-A DBMS includes data backup and recovery functions to ensure data availability at all times.

As potentially many users rely on the availability of the data, the data must be safeguarded against possible calamities. Regular back-ups of the data set, and automatic recovery schemes provide an insurance against loss of data.

-A DBMS allows the control of data redundancy.

A well-designed database takes care of storing single facts only once. Storing a fact multiple times—a phenomenon known as data redundancy can lead to situations in which stored facts may contradict each other, causing reduced usefulness of the data.

3. Write a note on Spatial Data functionality.

(Apr-19)

Ans:

- **Spatial Data functionality. (SDF)**

The main problem was that there is additional functionality needed by DBMS in order to process and manage spatial data. As the capabilities of our hardware to process information has increased, so too has the desire for better ways to represent and manage spatial data.

During the 1990's, object-oriented and object-relational data models were developed for just this purpose. These extend standard relational models with support for objects, including 'spatial' objects.

Currently, GIS software packages are able to store spatial data using a range of commercial and open source DBMSs such as Oracle, Informix, IBM DB2, Sybase, and PostgreSQL, with the help of spatial extensions. Some GIS software have integrated database 'engines', and therefore do not need these extensions.

ESRI's ArcGIS, for example, has the main components of the MS Access data-Spatial DBMS base software built-in. This means that the designer of a GIS application can choose whether to store the application data in the GIS or in the DBMS

- **GIS and spatial databases**

Spatial data can be stored in a special database column, known as the geometry column, (or feature or shape, depending on the specific software package), as shown in Figure This means GISs can rely fully on DBMS support for spatial data, making use of a DBMS for data query and storage (and multi-user support), and GIS for spatial functionality. Small-scale GIS applications may not require a multi-user capability and can be supported by spatial data support from a personal database.

In summary, spatial data functionality in GIS is fundamental to the system's capability to organize, analyze, and visualize geographic information. It empowers users to make informed decisions, solve complex problems, and better understand the relationships between various elements in the geographic space. Spatial data functionality in GIS is critical for decision-making in various domains, including urban planning, environmental management, transportation, public health, and natural resource management. It empowers users to analyze complex spatial relationships and make informed decisions based on geographic information.

4. Explain the relational data model using suitable example. (Apr-19)

OR

Explain the relational data model using suitable example. (Nov-19)

OR

Write a short note on relational data model. (Apr-23)

Ans:

A data model is a language that allows the definition of:

- 1 The structures that will be used to store the base data,
- 2 The integrity constraints that the stored data has to obey at all moments in time, and
- 3 The computer programs used to manipulate the data.

For the relational data model, the structures used to define the database are attributes, tuples and relations. Computer programs either perform data extraction from the database without altering it, in which case we call them queries, or they change the database contents, and we speak of updates or transactions.

Relations, tuples and attributes A table or relation is itself a collection of tuples (or records). In fact, each table is a collection of tuples that are similarly shaped. An attribute is a named field of a tuple, with which each tuple associates a value, the tuple's attribute value.

An attribute's domain is a (possibly infinite) set of atomic values such as the set of integer number values, the set of real number values, etc.

When a relation is created, we need to indicate what type of tuples it will store. This means that we must

- 1 Provide a name for the relation,
- 2 Indicate which attributes it will have, and
- 3 Set the domain of each attribute.

Finding tuples and building links between them

A key of a relation comprises one or more attributes. A value for these attributes uniquely identifies a tuple if we have a value for each of the key attributes, we are guaranteed to find no more than one tuple in the table with that combination of values. It remains possible that there is no tuple for the given combination.

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5. Differentiate between Vector data and Raster Data.

(Apr-19)(Nov-19)(Apr-23)

Ans:

The way that data is stored plays a central role in the processing and the eventual understanding of that data. In most of the available systems, spatial data is organized in layers by theme and/or scale.

In a GIS, features are represented with their (geometric and non-geometric) attributes and relationships. The geometry of features is represented with primitives of the respective dimension: a windmill probably as a point, an agricultural field as a polygon. Vector data types describe an object through its boundary, thus dividing the space into parts that are occupied by the respective objects. The raster approach subdivides space into (regular) cells, mostly as a square tessellation of dimension two or three. These cells are called either cells or pixels in 2D, and voxels in 3D. The data indicates for every cell which real world feature it covers, in case it represents a discrete field. In case of a continuous field, the cell holds a representative value for that field.

Sr.No	Vector	Raster
1.	Complex Data structure representation.	Simple data structure representation.
2.	Overlay more difficult of implement.	Overlay implementation in simple way.
3.	Inefficient in image processing as it works using function.	Efficient for image processing as it works with pixel.
4.	Topologies represented efficiently.	Difficult is representing topologies.
5.	Adapts well to scale changes.	Allow poor compactness of data while compression.
6.	Allow association of data and attribute.	Cell boundaries are independent of feature boundary.
7.	More update intensive.	
8.	Allow representing network.	

Raster Representation	Vector Representation
Advantages	
Simple data structure simple implementation of overlays efficient for image processing	Efficient representation of topology adapts well to scale changes allows representing networks allows easy association with attribute data
Disadvantages	
Less compact data structure difficulties in representing topology cell boundaries independent of feature boundaries	Complex data structure overlay more difficult to implement inefficient for image processing more update-intensive

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6. Write a note on Spatial Data Infrastructure.**(Apr-19)****Ans:**

Government, industry leaders, academics, and nongovernmental organizations connect each other with the analytic knowledge they need to make the critical decisions. Many organizations work in a cooperative setting in which geographic information is obtained from, and provided to, partner organizations and the general public. E.g Natural Earth Data allow downloading GIS data from <http://www.naturalearthdata.com/downloads/>, Open Government Data (OGD) Platform in India -

<https://data.gov.in/keywords/gis>. Typically, an SDI provides its users with different facilities for finding, viewing, downloading and processing data. Because the organizations in an SDI are normally widely distributed over space, computer networks are used as the means of communication. With the development of the internet, the functional components of GIS have been gradually become available as web-based applications.

Spatial Data Infrastructure (SDI) deals with the sharing of spatial data between the GIS in various organizations with the key importance and aspects of data dissemination, security, copyright and pricing requirement.

Spatial Data Infrastructure (SDI) is defined as "the relevant base collection of technologies, policies and institutional arrangements that facilitate the availability of and access to spatial data". Fundamental to those arrangements are the agreements between organizations and the agreements between software systems on how to share the geographic information. In SDI, standards are often the starting point for those agreements. Standards exist for all facets of GIS, ranging from data capture to data presentation. They are developed by different organizations, of which the most prominent are the International Organization for Standardization (ISO) and the Open Geospatial Consortium (OGC).

SDI provides users with different facilities for finding, viewing, downloading and processing data. Because the organizations in an SDI are normally widely distributed over space, computer networks are used as the means of communication. With the development of the internet, the functional components of GIS have been gradually become available as web-based applications. Much of the functionality is provided by so called geo-webservices, software programs that act as an intermediate between geographic data and the users of the web. Geo-webservices can vary from a simple map Data Management and Processing Systems display service to a service which involves complex spatial calculations. For their spatial data handling, these services commonly use standardized raster and vector representations.

In summary, Spatial Data Infrastructure is a critical component of GIS that provides the foundation for effective spatial data management, sharing, and collaboration. By promoting interoperability and standardization, SDI enhances the accessibility and usability of spatial information, contributing to informed decision-making and sustainable development.

7. Write a note on Spatial Data presentation.

(Nov-19)

Ans:

- **Spatial data presentation:**

The presentation of spatial data, whether in print or onscreen, in maps or in tabular displays, or as 'raw data', is closely related to the disciplines of cartography, printing and publishing. The presentation may either be an end-product, for example as a printed atlas, or an intermediate product, as in spatial data made available through the internet.

By Spatial data, we mean representations that can be operated upon by a computer. More specifically, by spatial data we mean data that contains positional values, such as (x;y) co-ordinates. Sometimes the more precise phrase geospatial data is used as a further refinement, which refers to spatial data that is georeferenced. The presentation of spatial data, whether in print or on- screen, in maps or in tabular displays, or as 'raw data', is closely related to the disciplines of cartography, printing and publishing.

Method	Devices
Hard Copy	Printer plotter (pen plotter, ink-jet, thermal transfer printer electrostatic plotter) film writer
Soft Copy	Computer screen
Outpt of Digital data sets	Magnetic tape CD-ROM or DVD the internet

In conclusion, spatial data presentation in GIS is a multidimensional process that involves careful consideration of design principles, symbology, visualization techniques, and the use of various media. Effectively presenting spatial data enhances communication, decision-making, and the overall understanding of complex geographic information.



8. What are the different ways of spatial data capture and preparation?

Explain.

(Nov-19)

Ans:

- **Spatial data capture and preparation:**

The functions for capturing data are closely related to the disciplines of surveying engineering, photogrammetry, remote sensing, and the processes of digitizing, i.e. the conversion of analogue data into digital representations. Remote sensing, in particular, is the field that provides photographs and images as the raw base data from which spatial data sets are derived.

Data can be collected through first hand observation called as primary source, or through individual, organization or published data called as secondary source. Capturing and acquisition is done through scanning, photogrammetric, remote sensing, digitization of analog map, field survey, GPS survey or manual data entry. Remote sensing, in particular, is the field that provides photographs and images as the raw base data from which spatial data sets are derived. Surveys of the study area often need to be conducted for data that cannot be obtained with remote sensing techniques, or to validate data thus obtained. Traditional techniques for obtaining spatial data, typically from paper sources, included manual digitizing and scanning. In recent years there has been a significant increase in the availability and sharing of digital (geospatial) data.

The data, once obtained in digital format, may not be quite ready for use in the system. This may be because the format obtained from the capturing process may not fit the format required for storage and further use, which means that some type of data conversion is required. This problem may also arise when the captured data represents only raw base data, out of which the real data objects of interest to the system still need to be constructed. For example, semi-automatic digitizing may produce line segments, while the application's requirements are that non-overlapping polygons are needed. A build-and-verification phase would then be needed to obtain these from the captured lines.

Method	Devices
Manual digitizing	coordinate entry via keyboard digitizing tablet with cursor mouse cursor on the computer monitor (heads-up digitizing) (digital) photogrammetry
Automatic digitizing	Scanner
Semi-automatic digitizing	line-following software
Input of available digital data	CD-ROM or DVD-ROM via computer network or internet (including geo-webservices)

In summary, spatial data capture and preparation involve a diverse set of methods, each serving specific purposes in the development and maintenance of GIS datasets. The choice of method depends on the nature of the data, the required accuracy, and the specific goals of the GIS project.

9. Define GIS. Explain its range of capabilities to handle georeferenced data.**(Apr-23)****Ans:**

GIS stands for Geographic Information System. It is a powerful tool used to capture, store, analyze, manage, and present spatial or geographic data. GIS combines hardware, software, and data to provide a comprehensive way to understand and visualize patterns, relationships, and trends in geographic information. Here's an explanation of GIS capabilities in handling georeferenced data:

- **Data Capture and Preparation:**

Definition: GIS allows the acquisition of geographic data through various methods, such as digitizing maps, using GPS devices, or remotely sensing the Earth's surface.

Capabilities: GIS tools enable the conversion of analog information (e.g., paper maps) into digital formats. This involves the creation of accurate and georeferenced datasets, which are essential for spatial analysis.

- **Data Management:**

Definition: GIS facilitates the organization and storage of geographic data in a structured manner.

Capabilities: GIS databases manage spatial data efficiently. It involves creating layers of information (e.g., roads, land use, elevation) that can be linked to attribute data (e.g., population, temperature). GIS manages relationships between different layers, ensuring data integrity and consistency.

- **Data Manipulation and Analysis:**

Definition: GIS allows the manipulation and analysis of spatial data to derive meaningful insights.

Capabilities: GIS provides tools for overlay analysis, spatial querying, and spatial modeling. Users can analyze relationships between different geographic features, perform spatial statistics, and generate models to simulate real-world scenarios. This capability is crucial for decision-making processes in various fields, such as urban planning, environmental management, and disaster response.

- **Data Presentation:**

Definition: GIS facilitates the visualization and communication of spatial information.

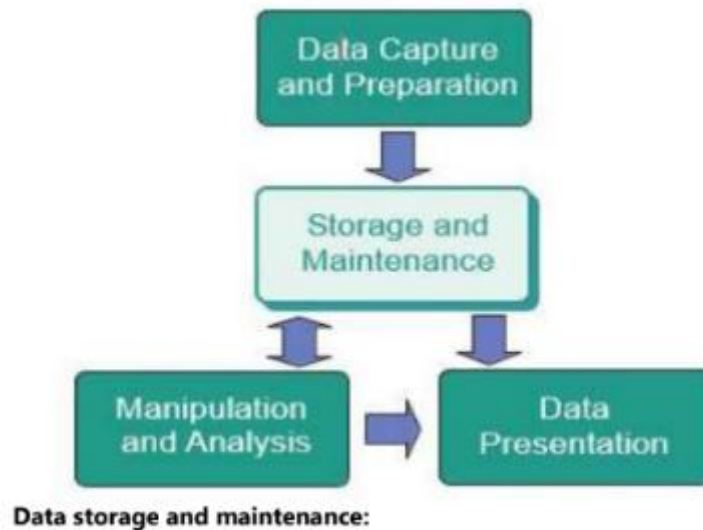
Capabilities: GIS produces maps and visualizations that aid in understanding complex spatial patterns. It allows users to create thematic maps, 3D visualizations, and interactive maps for effective communication. GIS outputs can be shared in various formats, such as printed maps, online maps, or interactive applications, making it accessible to a wide audience.

In summary, GIS is a versatile technology that encompasses a range of capabilities to handle georeferenced data, from initial data capture and preparation to sophisticated data analysis and presentation. Its applications are diverse, spanning fields such as environmental science, urban planning, natural resource management, public health, and more.

10. Explain the GIS Architecture and functionality using suitable diagram.

(Apr-23)

Ans:



• GIS Architecture and Functionality

Geographic information system consists of software, data, people, and an infrastructure. A GIS consists of several functional components-components which support key GIS functions. These are data capture and preparation, data storage, data analysis, and presentation of spatial data.

• Data capture and preparation

Data can be collected through first hand observation called as primary source, or through individual, organization or published data called as secondary source: Capturing and acquisition is done through scanning, photogrammetric, remote sensing, digitization of analog map, field survey, GPS survey or manual data entry. This data is then prepared for an application under consideration by removing error, calibration, checking quality, rasterization, vectorization etc.

• Data storage

The different types of information required for a GIS require storage which allows the information to be updated and queried for analysis by the user. Spatial data is usually stored as themes, layers, or coverage. Attribute data is the information about an object or feature.

• Data analysis

A good system and/or software package allows the user to define and execute spatial and attribute procedures. This is commonly thought of as the heart of the GIS. Overlaying, buffering, modeling, and analysis are some of the methods used in building a coverage or project.

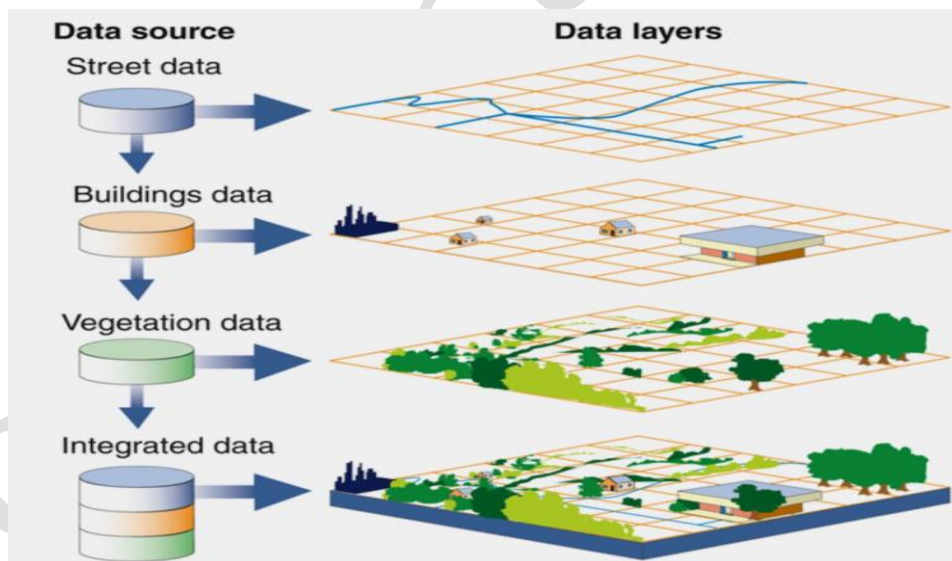
• Presentation of spatial data

After the careful preparation on several mapping tools which are integrated with GIS, the maps are presented to users. The final maps are of high cartographic quality and are brought out using a wide range of devices.

11. Explain the process of linking GIS and DBMS.**(Apr-23)****Ans:**

- **Linking GIS and DBMS**

GIS software provides support for spatial data and thematic or attribute data. GIS stores spatial data and attribute data separately. This required the GIS to provide a link between the spatial data (represented with rasters or vectors), and their non-spatial attribute data. The strength of GIS technology lies in its built-in 'understanding of geographic space and all functions that derive from this, for purposes such as storage, analysis, and map production. GIS packages themselves can store tabular data; however, they do not always provide a full-fledged query language to operate on the tables, DBMSs have a long tradition in handling attribute (ie. administrative, non-spatial, tabular, and thematic) data in a secure way, for multiple users at the same time. DBMS offer much better table functionality, since they are specifically designed for this purpose. A lot of the data in GIS applications is attribute data, so it made sense to use a DBMS for it. For this reason, many GIS applications have made use of external DBMS for data support. In this role. The DBMS serves as a centralized data repository for all users, while each user runs her/his own GIS software that obtains its data from the DBMS. This meant that a GIS had to link the spatial data represented with rasters or vectors, and the attribute data stored in an external DBMS. With raster representations, each raster cell stores a characteristic value. This value can be used to look up attribute data in an accompanying database table.



In vector representation, a spatial object like points, lines or polygon, are automatically given a unique identifier by the system. This identifier is usually just called the object ID or feature ID and is used to link the spatial object (as represented in vectors) with its attribute data in an attribute table. The principle applied here is similar to that in raster settings, but in this case each object has its own identifier. The ID in the vector system functions as a key, and any reference to an ID value in the attribute database is a foreign key reference to the vector system.

12. Explain spatial Database querying with suitable example.**Ans:**

Spatial database querying involves retrieving and manipulating data with a spatial component from a database that supports spatial data types and operations. The examples provided below assume the use of a spatial database extension like PostGIS for PostgreSQL, which enhances the database with spatial functionality. Spatial queries in GIS enable users to analyze and extract relevant information based on geographic relationships.

A Spatial DBMS provides support for geographic co-ordinate systems and transformations. It also provides storage of the relationships between features, including the creation and storage of topological relationships. As a result one is able to use functions for 'spatial query' (exploring spatial relationships). To illustrate, a spatial query using SQL to find all the Metro City within 20 km of a River GANGA would look like this:

```
SELECT C. Name  
FROM River AS R,  
City as C  
WHERE C. Type = "METRO" AND  
R. name = "GANGA" AND  
ST Intersects (C. Geometry, CT Buffer(R. Geometry, 20000))
```

In this case the WHERE clause uses the ST Intersects function to perform a spatial join between a 20000 m buffer of the selected River and the selected subset of Cities. The Geometry column carries the spatial data.

GIS spatial querying is essential for urban planning, environmental management, disaster response, and many other applications where understanding spatial relationships is crucial for decision-making.

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